Boiler Specifications
J-1502000-07

SECTION	TITLE
01010	General Requirements
15001	Repair of Boilers, Associated Services and Distribution Systems
Appendix J-1502000-02	References and Technical Documents (Separate Attachment)

SECTION 01010 GENERAL PARAGRAPHS

1. GENERAL DESCRIPTION: The Contractor shall provide all the necessary labor, equipment and materials and performance of all operations in connection with the following: Repairs and/or replacement of 6 psi steam valves through 900 psi steam valves; perform "R" stamp welding; With code required inspections and documentation resurfacing and welding of boiler hand-hole seats; refractory repair; boiler tube repair and replacement; 120 psi air system; 15 psi through 900 psi boiler water feed systems; steam and condensate piping systems; boiler certification; and other associated equipment and systems. This contract also contains provisions for the direct acquisition by the contractor of specialized sub-contracting personnel which may be required to address deficiencies not directly related to boilers but are nonetheless integral components of the power plant and its mission. The areas which may require sub contracted services include but may not necessarily be limited to the pure water facility, screen house, NPDES monitoring systems, oily waste water treatment facility and other miscellaneous facilities or utilities associated with the power plant.

2. SUBCONTRACTING: When requested by the government to acquire services outside of the contractors customary area of endeavor the Government will provide the Contractor with the following:

1. A brief general description of the required work or the nature of the situation which requires correction
2. Technical requirements and specifications for the intended work
3. A timeframe in which work must be completed
4. The standards to which the work must be performed to be considered acceptable

2.1 When requested by the government to acquire services outside of the contractors customary area of endeavor the Contractor shall:

1. Solicit proposals from three small business concerns that are experienced and capable of providing the required services.
2. Submit the lowest price proposal to the Government within two weeks from the date of the original request.

2.2 The Government will issue a delivery order in the amount corresponding to the accepted Contractor proposal indicated in the bid schedule. The Contractor is advised that all subcontracted work is subject to all of the insurance, wage and other miscellaneous requirements of this contract. The Contractor shall be responsible for all quality control associated with subcontracting activities. All subcontracted work is subject to Government acceptance and approval prior to payment.

3.0 In the event of emergency repairs, the Contractor shall be required to (a) arrive onsite within 2 hours after the Contractor receives a call from the Government, (b) submit a proposal within 24 hours after site visit, including time to complete, (c) commence work within 24 hours after the Contractor receives the notice to proceed from the Government, (d) prosecute the work diligently, and (e) compete the entire work ready for use not later than indicated on aforementioned proposal.

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SECTION

15001 PART

1 GENERAL

1.1 APPLICABLE PUBLICATIONS:

1.2 The publications listed below form a part of this specification to the extent referenced. REFERENCES AND TECHNICAL DOCUMENTS can be found in ATTACHMENT J-1502000-02.

1.3 SCOPE OF WORK: The work on any boilers includes repair or replacement of refractory brick, plastic refractory, castable refractory, and special shape refractory; boiler tubes, burners and generators, and applicable valves. Other work includes performing "R" stamp welding, welding and resurfacing boiler hand-hole seats, boiler certification, work on distribution systems, work on auxiliary equipment, work on piping, work in and around buildings 29, 490, 553, 554 and 318. Add Open and have boiler ready for inspection. Remove safety valves for overhaul and inspection and reinstall once inspection is complete.

1.4 QUALIFICATIONS OF CONTRACTORS: Organizations or companies engaged in the repair and alteration of high pressure steam boilers shall possess a State Unlimited Contractor License. Welders shall possess a National Board "R" Stamp for the repair of boilers and high pressure vessels. Requirements are included in the current edition of the National Board Inspection Code (NBIC), Part 3, Section 1. Boilermakers shall possess a State S-2 Unlimited Journeyman License. Welders and Boilermakers employed at the work site shall be certified by the Contractor as fully qualified journeyman. Journeymen helpers shall be trade experienced.

1.5 All repair work shall be in accordance with the following:

(1) Manufacturer's Instructions

1.5.1 REPAIR AND REPLACEMENT PARTS: The Contractor may be required to provide new parts and components when performing maintenance/repair services. All parts shall carry a 1 year warranty from the date of installment. All parts shall equal or exceed the quality of the original part being replaced. The Contractor shall retain original parts for at least fourteen (14) days following completion of the job and make these parts readily available for inspection by the Officer in Charge upon request.

1.5.2 LATENT DEFICIENCIES: The Contractor shall report to the Contracting Officer all boiler deficiencies discovered which were not indicated for repair.

1.5.3 DEFICIENCY ASSOCIATED WORK: The Contractor shall perform all necessary work required to get to the deficiency to make the required repairs. Work shall be in compliance with the procedures approved by the Contracting Officer.

1.6 SUBMITTALS:

1.6.1 SAMPLES: Samples shall be representative of the materials to be supplied and those upon which the specified tests have been made. The following samples shall be submitted to and approved by the Contracting Officer prior to delivery of materials to the project site. The Contractor shall submit material samples per Section F.

(1) Refractory Brick
(2) Special shape refractories
(3) Plastic Refractory
(4) Castable Refractory
(5) Refractory Mortar
(6) Refractory Anchors
(7)Miscellaneous Materials related to sub contracted work ordered by the Government shall be

submitted on an as required basis. Material submittal requirements will be spelled out in the request for proposal associated with the sub contracted work.

1.6.2 MANUFACTURE'S DATA: Submittals for each manufactured item shall be manufacturer's descriptive literature of cataloged products, equipment drawings, diagrams, performance and characteristic curves, and catalog cuts. The Contractor shall submit manufacturer's data per Section F.

1.6.3 CERTIFICATES OF CONFORMANCE OR COMPLIANCE: Submit for Contracting Officer approval, certificates from the manufacturer attesting that materials meet the requirements specified herein. The Contractor shall submit certificates of compliance per Section F.

1.7 DELIVERY - HANDLING - STORAGE: Materials shall be adequately packaged and protected during shipment and shall be inspected for damage upon delivery to the job site. Damaged materials that cannot be restored to like new condition shall be removed from the site and replaced at no additional cost to the Government.

2. PRODUCTS: Boiler and Related Equipment Inventory can be found in J-1502000-09

2.1 REFRACTORIES:

2.1.1 REFRACTORY BRICK: shall be high heat duty fire clay brick used in utility boilers; be composed of heat resistant materials of low heat conductivity; have shape and dimensions to match existing (floor is two layers of 2-1/2" thick brick, sidewall is 3" thick brick) or other approved dimensions; be resistant to thermal shock; be of homogeneous structure, free from warpage, cracks, laminations, segregations, void defects or soft centers. Fire brick shall conform to ASTM Specification C27-98, spall resistant, high duty, with a minimum pyrometric cone equivalent (PCE) of 31-1/2.

2.1.2 SPECIAL SHAPE REFRACTORIES: For bull ring and burner ring tiles shall be of substantially the same quality and rating as the refractory brick and shall accurately fit the dimensions and anchorage of the boiler. Plastic or cast refractory will not be acceptable as a substitute for special shapes. Dimensions and shapes shall be as specifically recommended by the boiler manufacturer.

2.1.3 PLASTIC REFRACTORY: Shall have high fusion, resistance to spalling, a minimum of shrinkage, and be easily workable. Plastic refractory shall be high duty fire clay conforming to ASTM C27-84 with a pyrometric cone equivalent (PCE) of 31 to 32. Specification C401-84, Class F.

2.1.4 CASTABLE REFRACTORY: Shall be hydraulic setting aluminum-silicon-base, service temperature of 3,100 degrees F. conforming to ASTM Specification C401-84, Class F. Castable refractory shall be suitable for casting or ramming.

2.1.5 REFRACTORY MORTAR: Shall be air setting bonding mortar of super duty ire clay quality with a service temperature of 3,100 degrees F. and pyrometric cone equivalent (PCE) of 34.

2.7 VALVES: Shall be rated as applicable to the system requiring repair or replacement. Valves need to be installed IAW code requirements that may be but not limited to either being flanged or welded to piping. Valves shall conform to MIL-V-16733D, MIL-V-18436F, MIL-V-18634B. Safety relief valves must be repaired by a certified safety valve repair shop or by the valve manufacturer that can provide required supporting documentation for the repair or replacement. A "V" stamped facility is preferred.

3. INSTALLATION

CONSTRUCTION: Construction shall conform as close to details of the manufacturer's reference drawings for the refractory lining as practical and to the drawing details and to details of the existing refractory lining and piping as practical. Deviations from manufacturer's reference drawings must be approved by NAVFAC UEM Engineering prior to proceeding. Contractor shall submit Deviations from Manufacture's Design specs shall be per Section F. Special attention shall be given to providing for expansion and to the prevention of future damage by expansion and slag intrusion. All work shall be accomplished by experienced tradesmen in accordance with standard industry practice. Two copies of manufacturer's data must be supplied with equipment installed, one copy to building 29 Facility Manager and one copy to the Contracting Officer.

3.1.1 BRICKWORK: Brickwork in floors shall be laid dry and tightly butted and shall have a smooth, even surface. The top layer shall be sealed with refractory mortar to prevent entry of slag. The perimeter of the floor shall have a 2" wide expansion joint on all sides two courses deep. The expansion joint shall be filled with compressible insulation.

3.1.2 BRICKWORK: Brickwork in walls and burner cone shall be set in refractory mortar and shall be accurately placed, level, plump and closely fitted to prevent entry of slag. Burner tiles shall be installed to the dimensions and with anchorage as recommended by the boiler manufacturer and according to existing construction details. Brickwork shall be cut to fit around horizontal tie bars. Surfaces and joints of new brickwork shall be sealed with refractory mortar.

3.1.3 SLAG DAM: Slag dam shall be provided to prevent slag from entering the expansion joint at the edge of the floor. The slag dam shall be set on 1/2" thick, high temperature wool compressed to 1/4" thickness to prevent bonding to the floor. Slag dam shall be corbelled refractory brick 7-1/2" high sloped to drain away from walls and shall be covered with plastic refractory or castable refractory to seal all joints.
3.1.4 BURNERS: The Contractor shall remove and reinstallation the burners in accordance with manufacturer's printed instructions.

3.1.5 PLASTIC AND CASTABLE REFRACTORY: Shall be provided and worked into place so as to seal off the escape of combustion gases and reduce heat loss to a minimum. Materials shall be mixed and placed in accordance with the manufacturer's printed instructions and shall be well compacted to the required shapes without voids or weak spots.

3.1.6 REFRACTORY MORTAR: Shall be mixed according to the manufacturer's printed instructions for dipping or troweling.

3.1.7 CURING OF NEW REFRACTORY MATERIALS: Initial firing of the boiler shall be coordinated with the Contracting Officer. Firing of the boiler will be done by the Government.

3.1.8 ACCESS: Boiler components removed for access shall be handled carefully and stored so as to prevent damage. Upon completion of work, the removed components shall be reinstalled with new gaskets where applicable.

3.1.9 VALVE REPAIR WORK: Shall be accomplished to restore valve(s) to good, safe working condition. Removal of a flanged valve requires new matching gaskets at reinstallation. Removal and reinstallation of valves which are welded to pipe shall be in accordance with MIL-STD-22. All replacement parts used internally and externally shall be matched to existing styles and types. Before any valve is removed from pipeline, ensure that valve is secured safely from system pressure. This may involve temporary boiler shutdown. Reinstalled valves shall be free of any leakage.

3.1.10 WELDING: Conform to requirements of ASME Boiler and Pressure Vessel Code and American Welding Society's "Structural Welding Code" using electrodes and processes in conformance with applicable Tables of those requirements. All welding, both shop and field, shall be performed only by certified welders as qualified by tests per AWS or ASME. Shop and field welds shall be subject to approval by an Engineer and independent testing laboratory engaged by the Government. Groove welds, where required, shall be full penetration welds. Welding of pressure piping shall be done by a certified welder who has or whose employer has an ASME "R" stamp. Boiler piping work shall conform to the requirement of ANSI/NB-23 National Inspection Code.

4 SERVICE REPAIR WORK:

GENERAL: The work may include any type of boilermaker and/or welding repairs on any power plant associated equipment or facility including boilers, steam valves, refractory, boiler tubes, auxiliary equipment, and distribution systems. Repairs shall be accomplished in compliance with the References in J-1502000-02.

4.1 REDUCTION OF SERVICE OUTAGE: To expedite the repairs and minimize down time, the Contractor shall perform as many procedures and set up as much equipment as safely possible while the boiler is in service.

4.2 PROTECTION OF EXISTING EQUIPMENT: The Contractor shall observe all the necessary precautions to protect the boiler equipment, including auxiliaries, from damage preceding and during the repair work and cleanup.

4.3 DEFICIENCY ASSOCIATED WORK: The Contractor shall perform all the necessary work required to get to the deficiency to make the required repairs. Work shall be in compliance with the procedures approved by the Contracting Officer.

4.4 LATENT DEFICIENCIES: The Contractor shall report to the Contracting Officer all boiler deficiencies which were not indicated for repairs.

4.5 CLEANING: Upon completion of repair work, the Contractor shall remove all tools, material scraps, dirt and debris, generated as a result of the work. All material and equipment not to be salvaged and debris shall be removed from Government property and disposed of properly.

4.6 INDEFINITE QUANTITY WORK: The Contractor shall notify the OIC and provide a labor hour estimate for all work. The Government will negotiate labor hours with the Contractor by comparing the Contractor's proposal with accepted industry estimating standards such as RS Means, or others. If a mutually agreed upon labor hour figure cannot be agreed upon by both parties, the Government has the right to halt work and to complete the work by other means. For emergencies, the Contractor shall arrest the emergency conditions prior to leaving the job site. Additional compensation will be authorized only for those amounts which exceed the labor or material limitations.

4.7 SCAFFOLDING: Approximately 50% of refractory repair will require scaffolding.

5 CERTIFICATION AND TESTING:

5.1 QUALIFICATIONS OF INSPECTORS: All inspectors shall possess a valid certificate of competency issued by any political subdivision (such as state, province, territory, county or city) of the United States that has adopted the Boiler Pressure Vessel Code of the American Society of Mechanical Engineers, or a commission issued by the National Board of Boiler and Pressure Vessel Inspectors and other certification approved by the Naval Facilities Engineering Command.

5.2 INSPECTIONS AND TESTS: The Contractor shall provide qualified authorized inspectors to perform inspections of any code required work and provide the necessary supporting documentation. The internal and external inspections, inspection under hydro-static pressure operational inspection will be conducted by the navy authorized inspector. Emission stack testing of boilers will be required when due. The inspector will not physically operate any utility plant equipment or appurtenances but will witness these operations as performed by plant personnel. All inspections shall be performed in accordance with NAVFAC UFC-3-430-07 (INSPECTION AND CERTIFICATION OF BOILERS AND UNFIRED PRESSURE VESSELS) and with Section 22a-174-22(k) of the Regulations of Connecticut State Agencies Inspection requirements outlined under Section Vl (Maximum Allowable Working Pressure) and Section Vll (Pneumatic Tests) are not required for certification and will not be performed under this contract. Inspections may be conducted on three (3) separate days with a sufficient time period between days to allow for required boiler preparations. The time between inspection days and the sequence of inspection will be at the discretion of the Government. Boilers shall be tested on both natural gas and
#2 fuel oil. All applicable ASME and Nation Board codes etc. for the installation, repair and operation must be met.

5.3 INTERNAL AND EXTERNAL INSPECTIONS: Inspections shall be performed in one day and shall be in accordance with the inspection procedure outlined in UFC-3-430-07. The work shall include external inspection of the economizer tubes.

5.4 HYDROSTATIC TEST OF THE BOILER: Test will be performed in one day and shall be witnessed by the inspector. A close visual examination shall be conducted at the maximum allowable pressure accordance with code requirements. The boiler will be inspected under pressure as described in NAVFAC UFC-3-430-07.

5.5 OPERATIONAL TESTS: Test shall be made in one day. The inspector shall witness the operational test of operating controls as required by UFC-3-430-07.

5.6 EMISSIONS TESTING: Test will be performed in accordance with Section 22a-174-22(k) of the Regulations of Connecticut State Agencies Inspection requirements. Tests shall be performed in one day and shall be witnessed by the State of Connecticut. The Contractor shall schedule the test Six, (6) months prior to the due date of the test. The Contractor shall be responsible for submitting the required Intent to Test (ITT) package and copy of the test protocol for approval to the Connecticut Department of Environmental Protection. Copies shall be submitted to SUBASENLON. The Contractor shall also be responsible for notification to CTDEEP of testing date and arrange for appropriate attendance of CTDEEP personnel. The Contractor shall be responsible for all State fees for testing observation. The Contractor shall be responsible for submitting an acceptable test report to CTDEEP within 45 days of testing. The Turbine Generator shall be tested alone on natural gas and together with the HRSG Duct Burner on natural gas and #2 fuel oil. For all units testing shall consist of 3, 1-hour tests with the units operating at or above 90% maximum capacity.

5.7 REPORTS: The inspector will prepare an Inspection Report Form for Boilers, NAVFAC Form 9-11014/38 and will be submitted to the COR and the Utilities Division Maintenance Branch Foreman at the end of each working day. The report will contain any recommendations resulting from the day's inspection and shall list any repairs or modifications that will be required to obtain boiler certification.

5.8 CERTIFICATION OF BOILERS: The Contractor shall adhere to the rules and regulations as outlined in NAVFAC UFC-3-430-07. The certificate to be issued will be NAVFAC 11014/32, Inspection Certificate-Boiler/Unfired Pressure Vessel. The certificate will be issued by a naval boiler inspector. Certification shall include, but not limited to:

Open Steam and Mud Drums
-Remove Steam Drum Internals
-Clean Steam and Mud Drums For inspection
-Reinstall internals when cleaning and inspections are complete
-Close fire side access doors when cleaning and inspections are complete
-Clean Manway Seats, notify government if machining is required
-Close Steam and Mud Drums. Provide new gaskets for hatches, valves, or handholes where needed
-Depending on unit, every 10th handhole to be removed for inspection and re-installed after inspection
-Inspect in Soot Passages, Economizer, Fire Box. Notify government if problems exist
- Remove Drum, Economizer and Superheater Safeties, Reinstall prior to boiler pressure test

5.9 SCHEDULING OF WORK: The exact dates for providing service will be established between the Contractor and the Project Manager. The Contractor must schedule work with the government within five working days of receiving a notice to proceed. Once a schedule of performing work has been established and prior to starting any work, the Contractor will contact the Performance Assessment Representative. All schedules of work must be submitted to the contact Project Manager for approval prior to starting any work.

7 SAFETY

7.1 HOT WORK PERMIT: As required per task order the Contractor shall be responsible for obtaining a Hot Work Permit from the Subase Fire Department, (860) 694-3333 or (860) 694-3466 prior to any burning, brazing or welding.

7.2.1
7.22 WEAPONS ARC TRAINING: The Contractor shall attend the required Weapons Arc training as required per NAVSEA OP-5 and per task order.

7.23 RADIOLOGICAL ACCIDENT (RADCON) DRILLS: RADCON drills are conducted about once every three months. Traveling thought the Subase is prohibited throughout the duration of the drill. The drills last from two to four hours. All personnel on the Base shall listen to the public address system for specific instructions. All costs associated with delays due to RADCON drills are the responsibility of the Contractor.

8. WARRANTY MANAGEMENT: Prior to performing repair work, the Contractor shall report to the KO/COR all defects in workmanship, material, parts, or improper installation and found by the Contractor to be covered by a warranty. The Contractor is responsible for knowing which equipment and components are covered by the original warranty and the warranty duration. The KO/COR will provide available warranty documents.